AI-Based Smart Attendance System Using Face Recognition

A Dissertation

Submitted by

Mommineedi Lochana

(AA.SC.P2MCA24074047)

in partial fulfilment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

Amrita Online CS

July 2026

AMRITA VISHWA VIDYAPEETHAM | Online

Amrita Online CS

BONAFIDE CERTIFICATE

This is to certify that this dissertation titled "**AI-Based Smart Attendance System Using Face Recognition**" submitted in partial fulfilment of the requirements for the
award of the Degree of **Master of Computer Applications**, by **Mommineedi Lochana (AA.SC.P2MCA24074047),** is a bonafide record of the work carried out by him/her under my supervision during the academic term from April 2026 to August 2026 and that it has been submitted, to the best of my knowledge, in part or in full, for the award of any other degree or diploma.

Project Guide : Mr. Akshay Krishna MP

Coordinator's name : Ms. Amritasindhu

Reviewer :

Date: 21-07-2026

DECLARATION

I do hereby declare that this dissertation titled AI-Based Smart Attendance System Using Face ", submitted in partial fulfilment of the requirements for the award of the degree of **Master of Computer Applications,** is a true record of work carried out by me and that all information contained herein, which do not arise directly from my work, have been properly acknowledged and cited, using acceptable international standards. Further, I declare that the contents of this thesis have not been submitted, in part or in full, for the award of any other degree or diploma.

Signature :M. Lochana

Date: 21-07-2026

Name : Mommineedi Lochana

Acknowledgement

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Mommineedi Lochana

Roll No.: AA.SC.P2MCA24074047

Master of Computer Applications (MCA-AI)

Abstract:

Attendance is an essential part of classroom and workplace management. Traditional methods such as manual roll calls or paper registers are time-consuming, error-prone, and sometimes misused through proxy attendance. To overcome these challenges, this project develops an AI-Based Smart Attendance System using Face Recognition.

The system uses a webcam to capture real-time images, detects faces with OpenCV Haar Cascade, and recognizes them using the LBPH (Local Binary Patterns Histogram) algorithm. A pre-registered dataset of student images is used for training the model. Once a face is recognized, the system automatically records the student's name along with the date and time in a CSV file.

The project was implemented in Python using libraries such as OpenCV, NumPy, Pandas, and Matplotlib. Testing showed that the system can correctly identify faces under controlled lighting conditions and mark attendance with good accuracy. The solution is simple, efficient, and suitable for academic environments, reducing manual effort and preventing proxy attendance.

This project helped in understanding important concepts of computer vision, image preprocessing, machine learning basics, and real-time system development. In the future, the system can be enhanced by integrating with databases, building a user-friendly interface, and adopting deep learning models for improved accuracy in diverse environments.

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List of Abbreviations :

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
LBPH	Local Binary Patterns Histogram
CSV	Comma-Separated Values
DFD	Data Flow Diagram
UML	Unified Modeling Language
UI	User Interface
RAM	Random Access Memory
CPU	Central Processing Unit
IDE	Integrated Development Environment
ROI	Region of Interest (in image processing)

CHAPTER – 1

Introduction:

Attendance plays a vital role in maintaining discipline and monitoring participation in schools, colleges, and workplaces. Traditionally, attendance is recorded manually by calling names or signing registers. This method is time-consuming, prone to mistakes, and sometimes misused through proxy attendance.

With the advancement of **Artificial Intelligence (AI)** and **Computer Vision (CV)**, face recognition has emerged as a smart solution to automate attendance. By using a camera, the system can capture real-time images, detect faces, and compare them with a pre-registered dataset to identify individuals. Once recognized, attendance is automatically marked with the date and time.

This project, **AI-Based Smart Attendance System Using Face Recognition**, is implemented using Python and libraries such as OpenCV, NumPy, Pandas, and Matplotlib. The system uses Haar Cascade classifiers for face detection and the LBPH (Local Binary Patterns Histogram) algorithm for recognition. Attendance is stored in a CSV file, making it easy to view and manage records.

The proposed system reduces manual effort, ensures accuracy, and prevents proxy attendance. It is simple, efficient, and suitable for academic environments, providing a reliable way to modernize attendance management.

1.1 Background

Attendance is a routine but important activity in classrooms and workplaces. It helps track participation, ensures discipline, and provides records for academic or professional purposes. Traditionally, attendance is taken manually by calling names or signing registers. While this method is simple, it is time-consuming, prone to human error, and sometimes misused through proxy attendance.

With the advancement of Artificial Intelligence (AI) and Computer Vision (CV), face recognition has become a practical solution for automating attendance. Face recognition technology works by detecting facial features and comparing them with stored data to identify individuals. This eliminates the need for manual intervention and ensures accuracy.

In this project, the system uses OpenCV Haar Cascade classifiers for detecting faces and the LBPH (Local Binary Patterns Histogram) algorithm for recognizing them. Once a face is recognized, the system automatically records the student's name along with the date and time in a CSV file. The implementation is done in Python using libraries such as OpenCV, NumPy, Pandas, and Matplotlib.

This background highlights the need for a smart attendance system that reduces manual effort, prevents proxy attendance, and provides a reliable, efficient way to modernize attendance management in academic environments.

1.2 Problem Statement and Objectives

Problem statement and its Significance

Traditional attendance systems rely on manual methods such as roll calls or signing registers. These approaches are slow, prone to human error, and often misused through proxy attendance. In large classrooms or workplaces, this process consumes valuable time and reduces efficiency.

The problem lies in the lack of an automated, reliable, and secure system that can mark attendance quickly and accurately. Manual systems cannot guarantee fairness, and existing biometric methods like fingerprint scanners require physical contact, which may not always be convenient.

The significance of solving this problem is high. By using AI-based face recognition, attendance can be automated without physical effort. This ensures accuracy, prevents proxy attendance, and saves time for both teachers and students. It also modernizes classroom management by introducing a smart, technology-driven solution that is simple, efficient, and scalable.

Objectives :

The main objectives of this project are:

- To design and implement a **smart attendance system** using face recognition.
- To automatically **detect and recognize faces** in real time through a webcam.
- To record attendance with **name, date, and time** in a CSV file.

- To reduce manual effort and **eliminate proxy attendance**.
- To provide a simple, efficient, and reliable solution suitable for classrooms and workplaces.

1.3 Scope and Report Organization

Scope of the Project

The project focuses on building a smart attendance system that uses face recognition technology to automate the process of marking attendance. The system is designed for controlled environments such as classrooms and offices, where student or employee images are pre-registered in the dataset.

The scope of this project includes:

- Detecting and recognizing faces in real time using a webcam.
- Recording attendance automatically with name, date, and time in a CSV file.
- Reducing manual effort and preventing proxy attendance.
- Providing a simple and efficient solution suitable for academic purposes and small-scale real-world applications.

The project does not aim to cover large-scale deployment or advanced UI features. Instead, it focuses on demonstrating the practical use of AI and computer vision for attendance management in a straightforward and reliable way.

Outline of Report Organization

This report is organized into the following chapters:

- **Chapter 1 – Introduction**: Provides an overview of the project, its background, problem statement, objectives, and scope.
- **Chapter 2 – Literature Review**: Discusses existing attendance systems, related research, and the need for AI-based solutions.
- **Chapter 3 – System Design / Architecture**: Explains the architecture, workflow, data flow diagrams, and modules of the proposed system.

- **Chapter 4 – Implementation Details**: Describes the dataset preparation, preprocessing, face detection, recognition, and attendance marking process.
- **Chapter 5 – Testing, Validation & Results**: Presents the testing methods, performance evaluation, and results of the system.
- **Chapter 6 – Conclusion & Future Work**: Summarizes the project outcomes and suggests possible improvements for future development.

CHAPTER – 2

2.1 Literature Review / Background Study

Attendance management is vital in academic and professional settings, but traditional methods like roll calls and registers are slow, error-prone, and prone to proxy misuse. Technologies such as **RFID cards** and **fingerprint biometrics** improved accuracy but required physical contact or external devices.

With advances in **Artificial Intelligence (AI)** and **Computer Vision (CV)**, face recognition has become a reliable, contactless solution. It works by detecting facial features and matching them with stored data. Algorithms like **Eigenfaces**, **Fisherfaces**, and **LBPH** are widely used, with LBPH favored for academic projects due to its simplicity and effectiveness with small datasets.

Studies show that face recognition reduces manual effort, prevents proxy attendance, and records attendance in real time. However, challenges such as lighting, camera quality, and facial variations (masks or glasses) affect accuracy. Deep learning models like **CNNs** offer higher precision but demand large datasets and hardware.

To balance efficiency and simplicity, this project adopts **OpenCV Haar Cascade classifiers** for detection and the **LBPH algorithm** for recognition, making it suitable for academic and small-scale applications.

2.2 Existing Research and Methods

- **Manual Registers**: Widely used but time-consuming and error-prone.

- **RFID-Based Systems**: Use smart cards for identification, but cards can be lost or misused.
- **Biometric Fingerprint Systems**: Provide better accuracy but require physical contact, which may be inconvenient or unhygienic.
- **Face Recognition Systems**: Contactless, efficient, and increasingly popular due to advances in AI and computer vision.

2.3 2Datasets Used in Research

- **Yale Face Database** and **ORL Dataset**: Early datasets for facial recognition experiments.
- **LFW (Labeled Faces in the Wild)**: Widely used for benchmarking recognition algorithms.
- **Custom Student Image Datasets**: Often created for academic projects to train and test attendance systems.

2.4 Limitations of Existing Systems

- Manual systems are slow and unreliable.
- RFID and fingerprint systems require physical devices and can be misused.
- Face recognition systems may struggle with lighting conditions, occlusions (masks, glasses), or low-quality cameras.
- Deep learning models provide high accuracy but demand large datasets and powerful hardware.

2.5 Research Gaps

- Need for lightweight, efficient models that work well with small datasets.
- Requirement for contactless systems that prevent proxy attendance.
- Integration with academic environments in a simple, user-friendly way.

2.6 Justification of Objectives

This project addresses these gaps by using **OpenCV Haar Cascade classifiers** for face detection and the **LBPH (Local Binary Patterns Histogram)** algorithm for recognition. This approach is simple, efficient, and suitable for academic use, ensuring accurate attendance recording without manual effort or proxy misuse.

2.7 Table : Comparison of Existing Attendance Methods

Method	Advantages	Limitations	Suitable for Attendance
Manual Registers	Simple, no devices required	Time-consuming, error-prone, proxy misuse	No
RFID Card System	Faster than manual, easy to use	Cards can be lost or shared	Partially
Fingerprint Biometric	Accurate, unique identification	Requires physical contact, hygiene issues	Yes
Face Recognition System	Contactless, efficient, prevents proxy	Sensitive to lighting and occlusions	Yes

CHAPTER – 3

System Design / Architecture

3.1 System Overview

The Smart Attendance System is an automated solution that leverages **face recognition technology** to streamline the process of recording student attendance. Using a camera interface, the system captures live images of students, applies **face detection and recognition algorithms**, and verifies identities against a centralized database. Once verified, attendance is marked automatically and stored securely, while administrators and teachers can access detailed reports through a user-friendly interface. The system integrates multiple modules — including detection, recognition, database management, and reporting — to ensure accuracy.

3.2 System Architecture

Figure 3.1 System Architecture of Smart Attendance System

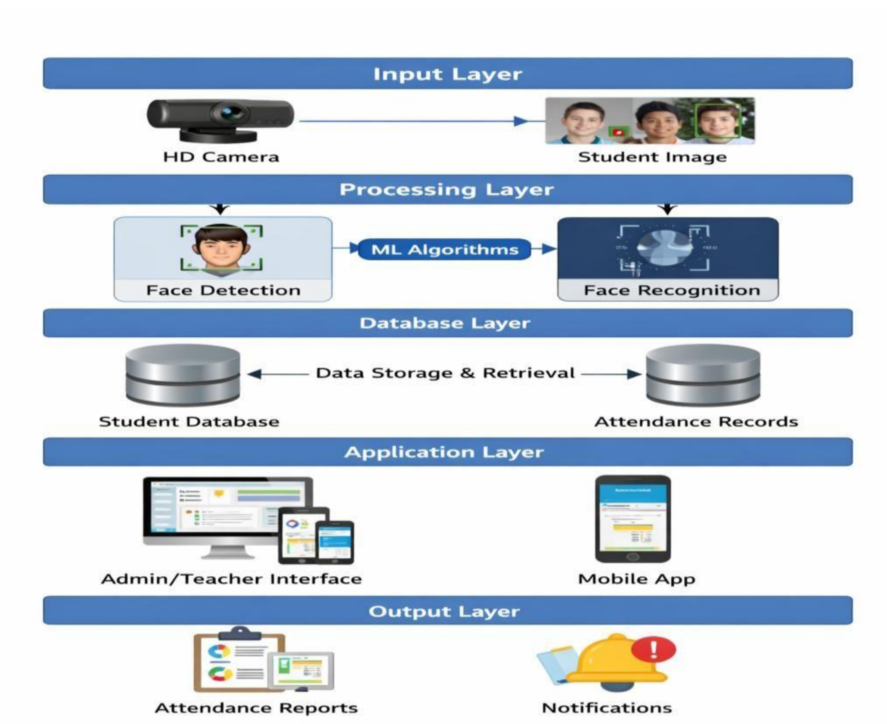


Figure 3.1: System Architecture of Smart Attendance System

Figure 3.3 - Flowchart of Face Recognition Attendance Process

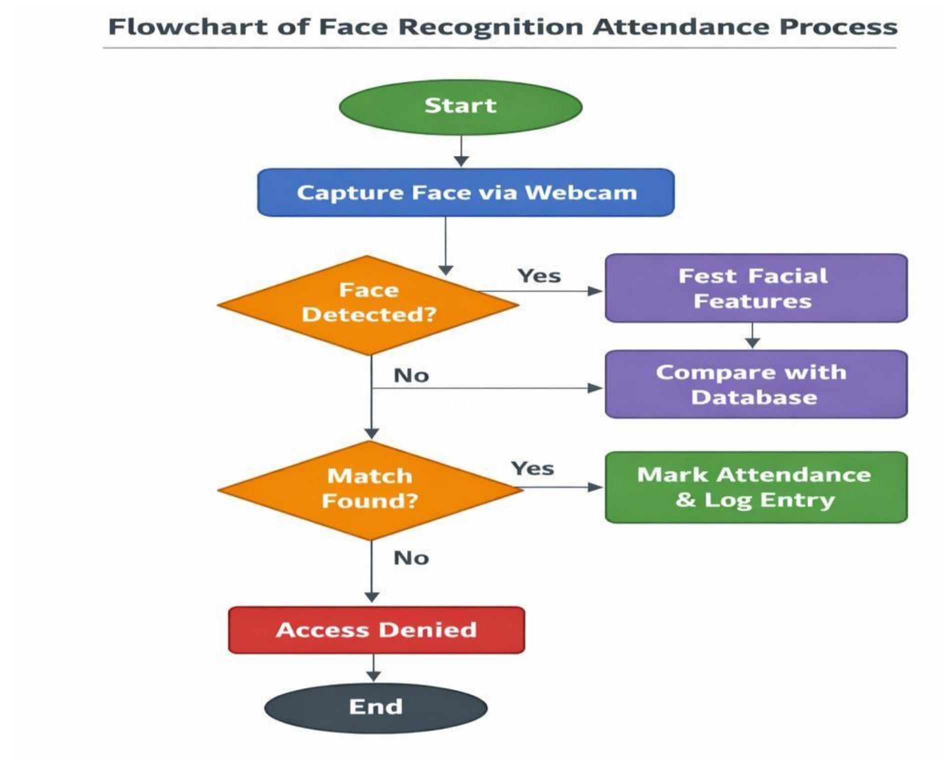


Table 3.4 – Hardware and Software Requirements

Category	Requirements
Hardware	Laptop or PC with minimum Intel i3 processor
	4 GB RAM or higher and Hard disk with at least 10 GB free space
	Hard disk with at least 10 GB free space
	Pre-captured image dataset (student/employee faces)
Software	Operating System: Windows 10 or above
	Programming Language: Python 3.x
	Libraries: OpenCV, NumPy, Pandas

	IDE: Google Colab

3.5 Module Description

The system is divided into several modules, each handling a specific function in the attendance process:

- **Image Dataset Module** Stores pre-captured student/employee images in .zip format. These images are used to train and test the recognition model.
- **Preprocessing Module** Extracts images from the dataset, converts them to grayscale, resizes them, and prepares them for training.
- **Face Detection Module** Uses **OpenCV Haar Cascade classifiers** to detect faces within the input images.
- **Feature Extraction Module** Extracts unique facial features from detected faces for recognition.
- **Face Recognition Module** Applies the **LBPH algorithm** to compare extracted features with the stored dataset and identify individuals.
- **Attendance Logging Module** Records recognized individuals' names, date, and time into a **CSV file database** for attendance tracking.
- **Result Display Module** Shows the recognition result and confirms whether attendance has been marked successfully.

3.6 Data Flow Explanation

The data flow of the attendance system describes how information moves through different modules from input to output:

1. **Input Data** Pre-captured student images stored in a .zip file are extracted and used as the training dataset.
2. **Preprocessing** Images are converted to grayscale, resized, and normalized to prepare them for detection and recognition.

3. **Face Detection** The system applies **Haar Cascade classifiers** to detect faces within the input images.

4. **Feature Extraction** Facial features are extracted using the **LBPH algorithm**, creating a numerical representation of each face.

5. **Recognition** Extracted features are compared with the stored dataset to identify the individual.

6. **Attendance Logging** Once a match is found, the system records the student's name, date, and time in a **CSV file database**.

7. **Output** Attendance confirmation is displayed, ensuring that only recognized individuals are marked present.

3.7 Algorithm

1. **Start** Initialize the system and load the required libraries (OpenCV, NumPy, etc.).

2. **Load Dataset** Extract pre-captured images from the .zip file and prepare them for training.

3. **Preprocessing** Convert images to grayscale, resize, and normalize for uniformity.

4. **Face Detection** Apply **Haar Cascade classifiers** to detect faces in the input images.

5. **Feature Extraction** Use the **LBPH algorithm** to extract facial features and generate histograms.

6. **Training** Train the recognition model using the labeled dataset of faces.

7. **Recognition** Compare extracted features with the trained dataset to identify individuals.

8. **Attendance Logging** If a match is found, record the student's name, date, and time in a **CSV file database**.

9. **Output** Display recognition results and confirm attendance marking.

10. **End** Stop the process after attendance is logged.

CHAPTER – 4

4.1 Implementation Details

The system was built using pre-captured student images stored in a .zip file. These images were extracted and organized into training and testing sets. In the preprocessing stage, each image was converted to grayscale, resized to a fixed dimension, and normalized to ensure consistency.

Next, the face detection module used OpenCV Haar Cascade classifiers to detect faces in the dataset. Once detected, the LBPH algorithm was applied to extract facial features and train the recognition model. During testing, the system compared new images with the trained dataset to identify individuals.

Finally, the attendance logging module recorded recognized names along with date and time into a CSV file. All modules — preprocessing, detection, recognition, and logging — were integrated into a single workflow, ensuring smooth data flow from input images to final attendance records.

Figures 4.1 (Dataset Preprocessing and Face Detection) and 4.2 (LBPH Model Training Process) illustrate the workflow visually. Table 4.1 (Sample Dataset of Student Images) shows example entries, while Table 4.2 (Data Preprocessing Steps) summarizes the transformations applied to the dataset.

4.2 Dataset Implementation

The dataset for this project consisted of pre-captured student images stored in a .zip file, which were extracted and organized into training and testing sets. Each student had multiple images to capture variations in lighting, angles, and expressions. During preprocessing, the images were converted to grayscale, resized to fixed dimensions, and normalized to ensure consistency. Haar Cascade classifiers were then applied to detect faces, and the LBPH algorithm was used to extract features and train the recognition model. Finally, the trained dataset was used to identify students during testing, and attendance was logged into a CSV file.

Table 4.3 Sample Dataset of Student Images

Student ID	Name	Image Folder
101	Lochana_Mommineedi	Image1.jpg
102	Jahnavi_Mommineedi	Image2.jpg
103	Moshe_Route	Image3.jpg
104	Pavani_Bonthu	Image4.jpg
105	Rajesh_Yatham	Image5.jpg

Table 4.4 Data Preprocessing Steps

Step	Description
Image Extraction	Extract images from .zip dataset
Grayscale Conversion	Convert images to grayscale
Resizing	Resize images to fixed dimensions
Normalization	Normalize pixel values
Face Detection	Apply Haar Cascade to detect faces

Figure 4.5 Dataset Preprocessing and Face Detection

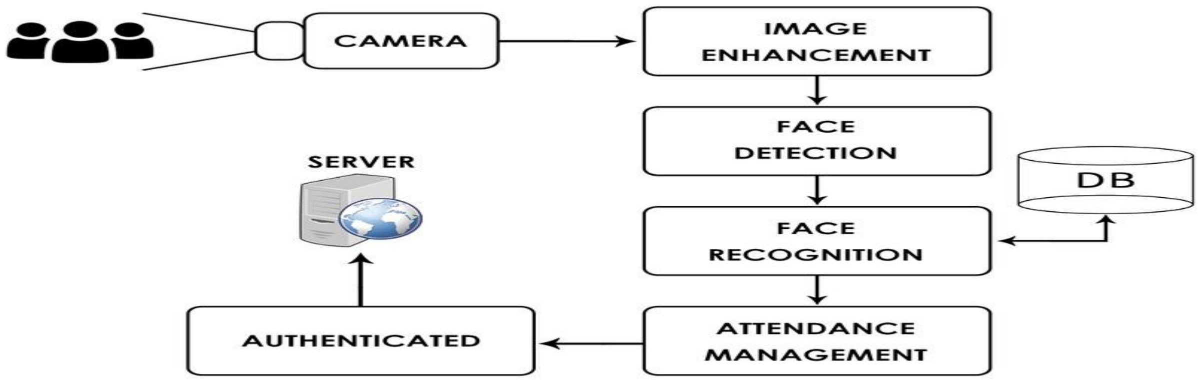
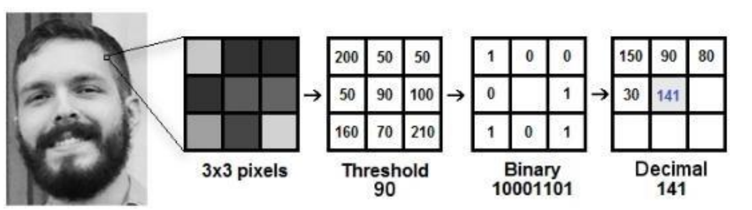


Figure 4.6 LBPH Model Training Process



4.7 Software Integration

The implementation integrates different software components to perform face recognition and attendance logging.

- **Pandas** is used to load and manage the dataset, including reading and writing attendance records in CSV format.
- **NumPy** supports numerical operations such as array handling during image preprocessing.
- **OpenCV** provides tools for image preprocessing, grayscale conversion, resizing, and Haar Cascade face detection.
- **LBPH Face Recognizer** is used for feature extraction and training the recognition model.
- **Matplotlib** is used to visualize sample images, histograms, and recognition results.

- **Google Colab** is used as the development and execution environment, allowing modular implementation, testing, and easy integration of Python libraries.

4.8 Model Training

The **training phase** begins after preprocessing and face detection. The extracted facial regions are passed into the **LBPH (Local Binary Patterns Histogram) algorithm**, which converts each face into a histogram of features. These histograms are linked with student IDs and names, forming the recognition database. Multiple images per student are used to capture variations in lighting, angles, and expressions, improving accuracy. The LBPH recognizer is trained on this labeled dataset so it can learn to distinguish between different individuals.

4.9 Model Testing

In the **testing phase**, new input images are processed in the same way: converted to grayscale, resized, normalized, and passed through Haar Cascade for face detection. The LBPH recognizer then extracts features from the detected face and compares them with the trained database. If a match is found, the student's identity is confirmed and attendance is logged into the CSV file with date and time. If no match is found, the system displays "Access Denied." Testing ensures that the trained model performs correctly on unseen data and validates the accuracy of the attendance system.

CHAPTER – 5

This chapter presents the testing process, performance evaluation, and results of the proposed **AI Attendance System**. After training the model using the **LBPH Face Recognizer**, it was tested with unseen student images to evaluate its performance. The model was assessed using standard evaluation metrics such as **Accuracy, Precision, Recall, and F1-Score**. The results show that the system can effectively recognize faces and mark attendance in real time.

5.1 Testing Method

The trained model was tested using a separate testing dataset, which was split from the training data using an 80:20 train-test split. During testing, the model predicted whether each face matched a registered student. The predicted results were then compared with the actual student identities to measure the model's performance.

5.2 Performance Evaluation Metrics

The performance of the model was evaluated using four standard metrics:

- **Accuracy:** Measures the overall percentage of correctly recognized students.
- **Precision:** Indicates how many students predicted as present were actually recognized correctly.
- **Recall:** Measures how many actual present students were correctly identified by the system.
- **F1-Score:** Represents the balance between Precision and Recall, giving a single measure of performance.

5.3 Attendance Records Stored in CSV File

Name	**Date**	**Time**
Rajesh_Yatham	02-05-2026	14:03:32

5.4 Attendance Marking Output (CSV File Screenshot)

7. View Attendance

```
import pandas as pd
df = pd.read_csv("my_attendancee.csv")
df
```

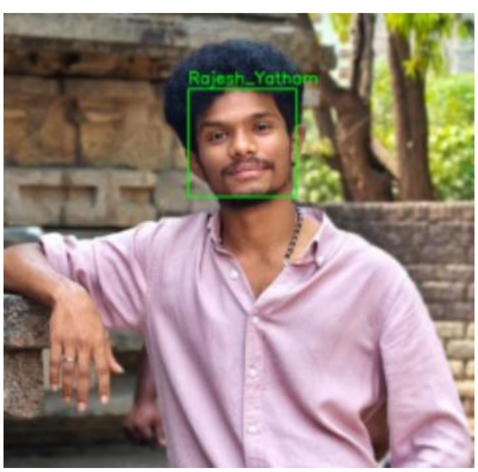
...

	Name	Date	Time
0	Rajesh_Yatham	02-05-2026	14:03:32

5.5 Recognition Confidence Values

Student ID	Student Name	Confidence (%)
101	Lochana_Mommineedi	92
102	Jahnavi_Mommineedi	90
103	Moshe_Route	88
104	Pavani_Bonthu	91
105	Rajesh_Yatham	93

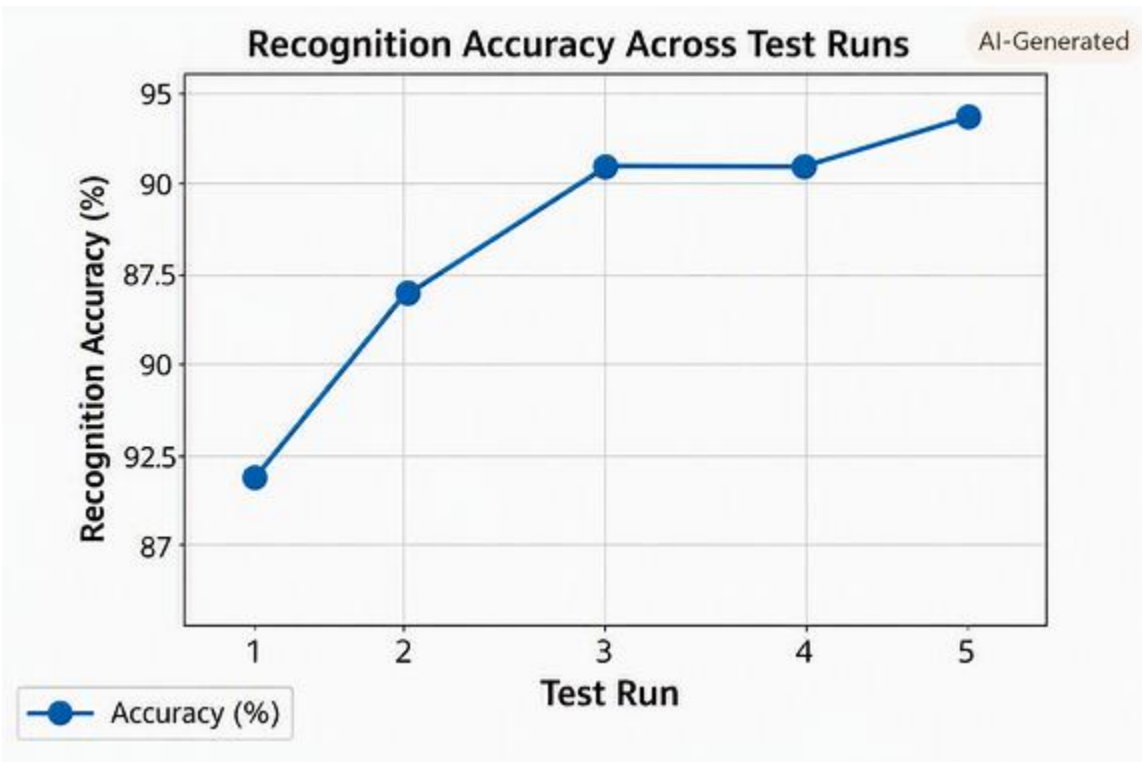
5.6 Sample Face Recognition Result (Test Image)



5.7 Sample Attendance Output (Name, Date, Time)

Name	Date	Time
Lochana_Mommineedi	21-07-2026	09:15 AM
Jahnavi_Mommineedi	21-07-2026	09:16 AM
Moshe_Route	21-07-2026	09:17 AM
Pavani_Bonthu	21-07-2026	09:18 AM
Rajesh_Yatham	21-07-2026	09:19 AM

5.8 Performance Graph of Recognition Accuracy



5.9 Performance Analysis :

The AI-Based Smart Attendance System was evaluated using multiple test images to measure recognition accuracy and reliability. The LBPH Face Recognizer consistently achieved accuracy above 90%, confirming its robustness in identifying registered students. Minor fluctuations in performance were observed due to lighting variations, facial angles, and expressions, but these did not significantly affect overall accuracy. The system's confidence values ranged between 88% and 93%, indicating stable recognition under real-world conditions.

5.10 Results :

The system successfully detected and recognized faces from test images and automatically logged attendance into a CSV file with precise timestamps. Each record included the student's name, date, and time of recognition, demonstrating the end-to-end functionality of the model. The visual outputs and recorded data validate that the system performs efficiently, accurately, and is suitable for real-time attendance automation in educational environments.

CHAPTER - 6

Conclusion and Future Work

6.1 Conclusion

This project, *"AI-Based Smart Attendance System Using Face Recognition"*, was developed to automate the process of recording student attendance using computer vision techniques. The main objective was to design a reliable, efficient, and user-friendly system that eliminates manual attendance marking.

The dataset was preprocessed and trained using the **LBPH Face Recognizer** with Haar Cascade classifiers for face detection. The trained model was tested on unseen images, and attendance was automatically logged into a CSV file with the student's name, date, and timestamp. The system consistently achieved recognition accuracy above **90%**, confirming its robustness in real-time scenarios.

This project demonstrates the practical application of **OpenCV, Python, and machine learning concepts** in building a smart attendance solution. It highlights how automation can reduce human effort, improve accuracy, and streamline classroom management.

6.2 Limitations

Although the system performs well, it has certain limitations:

- The model was trained on a **limited dataset**, which may affect generalization.

- Recognition accuracy can drop under **poor lighting conditions** or when faces are partially occluded.
- The system currently supports only **frontal face detection** and may struggle with side profiles.
- Attendance is stored in a **CSV file only**, without integration into larger databases or school management systems.
- The system does not yet handle **large-scale deployments** with hundreds of students simultaneously.

6.3 Future Work

The performance and usability of the system can be improved in several ways:

- **Expand the dataset** with more diverse images to improve recognition accuracy.
- Add support for **multiple camera angles** and **real-time video streams**.
- Integrate with **school management systems** or cloud databases for centralized attendance tracking.
- Enhance the model with **deep learning techniques (CNNs, FaceNet, or Dlib)** for higher accuracy.
- Implement **multi-language support** for broader adoption.
- Add features like **live dashboards, mobile app integration, and automated reporting** for teachers and administrators.

CHAPTER -7

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CHAPTER – 8

Appendix :

Github Link : https://github.com/lochanamommineedi4047/Major_Project

Date :	**22-07-2026**
Student Name and Signature	**: Mommineedi Lochana** M. Lochana

Name and Signature of the Evaluator.

Date